

MULTI-AGENT AI FRAMEWORK FOR OPTIMAL MULTI-DRUG COMBINATION THERAPY IN BREAST CANCER

*Extending Rule-Based Kinase Redundancy Analysis with Graph Neural Networks, Knowledge-
Graph-Grounded Retrieval, and Multi-Agent LLM Reasoning*

A Proposed Research Extension to the tnbc-genomics-agent Pipeline

Bioinformatics · Multi-Agent AI · Graph Neural Networks · Precision Oncology

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1. Executive Summary

Triple-negative breast cancer (TNBC) and other breast cancer subtypes that lack effective single-target therapy remain heavily dependent on combination regimens to overcome pathway redundancy and adaptive resistance. The existing *tnbc-genomics-agent* pipeline addressed this problem with a rule-based redundancy score and a single retrospective LLM narrative layer. That approach is a valuable first step, but it ranks candidate drug pairs purely by *shared pathway membership*, not by predicted pharmacological synergy, toxicity, or real-world trial evidence — three factors that determine whether a combination is actually effective and safe.

This report proposes a new approach, the **Multi-Agent Combination Oncology Reasoning Engine (MC-ORE)**, which extends the existing four-layer pipeline into a five-layer system that combines **(1) a graph neural network (GNN) synergy-prediction model** trained on cancer-cell-line synergy data, **(2) a biomedical knowledge graph** linking kinases, drugs, pathways, and disease indications, **(3) a retrieval-augmented generation (RAG) evidence layer** indexing PubMed and clinical-trial literature, and **(4) a multi-agent large language model (LLM) reasoning layer** in which specialist agents — genomics, pharmacology/synergy, clinical evidence, and toxicity/safety — debate and reconcile findings into an explainable, ranked set of combination recommendations, with an optional **(5) reinforcement-learning (RL) feedback layer** for longitudinal refinement.

The design directly responds to limitations the original pipeline documentation itself identifies (a 10-gene knowledge base, purely rule-based scoring, and single-agent LLM synthesis), and it is grounded throughout in peer-reviewed literature on drug-synergy prediction, multi-agent clinical LLM systems, and knowledge-graph-based drug repurposing in breast cancer.

2. Motivation: Limitations of the Current Rule-Based Approach

The *tnbc-genomics-agent* pipeline computes a pathway redundancy score from the number of altered kinase genes feeding a shared downstream pathway, using a fixed sigmoid-like formula, and then enumerates all co-affected kinase pairs sharing a pathway as “combination suggestions,” ranked only by how much literature evidence exists for each pair. This design has three structural limitations that motivate a new approach:

1. **Pathway overlap is a proxy for synergy, not a measurement of it.** Two kinases feeding the same pathway does not guarantee that co-inhibiting them produces a synergistic — rather than merely additive or even antagonistic — effect at the cellular level.
2. **Combination ranking ignores toxicity and pharmacokinetic interaction.** A pair with strong pathway rationale may still be clinically unusable because of overlapping organ toxicity or drug-drug interactions — information the current rule-based ranking does not model.
3. **A single LLM synthesis step has no mechanism for adversarial checking.** One reasoning pass over structured data cannot cross-examine its own conclusions against independent pharmacology or safety evidence, which limits how much clinicians can trust the ranked output without manual review.

Each of these gaps can be closed with AI techniques that have matured substantially since the original pipeline was designed: learned synergy-prediction models, knowledge-graph retrieval, and multi-agent LLM deliberation.

3. Related Work

3.1 Graph Neural Networks for Drug Synergy Prediction

Several recent models replace hand-crafted synergy heuristics with learned representations. **SynerGNet** builds cancer-specific featured graphs by mapping heterogeneous biological features onto the human protein–protein interaction (PPI) network and trains a GNN to classify drug pairs as synergistic; using AZ-DREAM Challenge data it reached a balanced accuracy of 0.68, improving to 0.73 with synthetic data augmentation, and generalised well on independent validation against DrugCombDB ^[1].

DeepDDS applies graph attention/convolutional networks (GAT/GCN) directly to drug molecular graphs derived from SMILES structures, combined with cell-line gene-expression embeddings from the Cancer Cell Line Encyclopedia, to predict synergistic combinations ^[2]. **SANEpools** adds a hierarchical, attention-based graph-pooling mechanism that not only predicts a synergy score but also surfaces the specific gene–gene and gene–drug sub-networks driving the prediction, directly addressing the interpretability gap of earlier black-box synergy models ^[3].

Complementary work has applied classical and deep machine learning to the O’Neil FDA-approved drug combination dataset to classify combinations as synergistic, additive, or antagonistic across multiple cancer types, and specifically flagged kinase-inhibitor pairings with mTOR, DNA-damage, or HDAC inhibitors as consistently promising multi-drug regimens ^[4] — a finding directly relevant to the kinase-centric design of the existing pipeline. More recent patient-derived-model approaches such as **PAIRWISE** extend this line of work toward personalised combination prediction rather than population-level synergy alone ^[5].

3.2 Multi-Agent LLM Systems and Retrieval-Augmented Generation in Oncology

A growing body of work shows that decomposing clinical reasoning across multiple specialised LLM agents, each grounded in retrieved evidence, outperforms a single general-purpose model. A multi-agent medical decision consensus system modelled an oncology multidisciplinary team (MDT) as seven role-specific LLM agents sharing a retrieval-augmented generation (RAG) module built on a vector database of clinical guidelines and roughly 34 million PubMed abstracts ^[6]. **EBMChat** implements an evidence-based-medicine workflow in which an agent iteratively converts clinical questions into queries, retrieves and evaluates evidence, and only then answers — explicitly addressing the tendency of single-pass RAG to ignore evidence timeliness and hierarchy ^[7].

An autonomous AI agent for oncology clinical decision-making, validated in *Nature Cancer*, argues that RAG alone “positions LLMs as mere information-extraction tools” and instead couples retrieval with reasoning, strategising, and tool use over patient records to reach true agentic decision support ^[8]. A 2025 survey of LLM-based multi-agent systems in medicine similarly identifies medical-knowledge RAG and external tool invocation — rather than a single fine-tuned model — as the dominant emerging design pattern for clinical-grade agents ^[9].

3.3 Knowledge Graphs and Drug Repurposing for Breast Cancer Specifically

DREAMwalk, published in *Nature Communications*, constructs a multi-layer biomedical knowledge graph spanning drugs, genes, and diseases and uses a semantic-information-guided random walk to predict drug–disease associations, improving link-prediction accuracy by up to 16.8% over prior state-of-the-art; the authors used breast carcinoma as one of their two validation case studies ^[10].

Most directly relevant, a 2024 case study applied **network medicine combined with generative AI** specifically to drug repurposing for breast cancer: the authors engineered a GPT-based model with few-shot prompting to mine clinical-trial text for combination-indicating phrases (e.g. “combined with,” “in addition to”), linked the resulting drug pairs to breast-cancer KEGG signalling pathways, and assembled a multilayer graph connecting drug combinations to their molecular targets ^[11]. This demonstrates, in the same disease context as the existing pipeline, that GenAI-assisted knowledge-graph construction is already a validated technique for surfacing candidate combination therapies. A broader 2025 review of AI-based drug repurposing in cancer confirms that knowledge-graph and deep-learning methods dominate repurposing, while generative and reinforcement-learning methods are increasingly used for de novo combination design ^[12].

3.4 Reinforcement Learning for Adaptive Treatment Optimisation

Beyond static ranking, reinforcement learning has been used to adapt combination and dosing decisions over time as patient state evolves. Deep RL controllers using model-free algorithms (Q-learning, normalized advantage functions, actor-critic methods) have been applied to jointly optimise chemotherapy and anti-angiogenic drug dosing, adjusting to patient-specific response with a reward function encoding tumour reduction and toxicity minimisation ^[13,14]. A dedicated review of RL strategies in cancer chemotherapy catalogues these approaches and identifies reward-function design and generalisation to real clinical data as the key open challenges ^[15] — directly informing the feedback-layer design proposed below.

4. Proposed Architecture: MC-ORE

MC-ORE (Multi-Agent Combination Oncology Reasoning Engine) preserves the existing pipeline's local-first privacy architecture and its four original layers, but replaces the rule-based redundancy analyzer and single-agent LLM synthesis step with three new AI-driven layers, and adds an optional adaptive feedback layer.

Layer	Component	Function	Relationship to Original Pipeline
L1	Multi-Omic Ingestion	Extends vcf_parser.py to optionally ingest RNA-seq expression and copy-number data alongside VCF variants, weighting kinase activation by transcriptomic evidence.	Direct extension (was Layer 1)
L2	Biomedical Knowledge Graph	Replaces the flat 10-gene kinase_db.py with a graph linking genes → pathways → drugs → trial evidence → disease indication, built from DGIdb, OncoKB, DrugCombDB, and ClinicalTrials.gov.	Replaces Layer 2
L3	GNN Synergy Prediction	A SynerGNet/DeepDDS-style graph neural network scores predicted synergy for every candidate drug pair (and higher-order combination) using PPI-network and cell-line features, replacing the sigmoid pathway-overlap heuristic.	Replaces Layer 3's scoring logic
L4	Multi-Agent LLM Reasoning	Specialist LLM agents (Genomics, Pharmacology/Synergy,	Replaces Layer 4 (single AI agent)

Layer	Component	Function	Relationship to Original Pipeline
		Clinical-Evidence/RAG, Toxicity & Safety, Chair) debate GNN output plus retrieved literature to produce a ranked, explainable recommendation set.	
L5	Adaptive Feedback (optional)	An RL module that re-ranks future recommendations using anonymised clinician acceptance/outcome signals as reward, following RL-for-treatment-optimisation literature.	New capability, no prior analogue

4.1 Layer 2 — Biomedical Knowledge Graph

The flat kinase database is restructured as a heterogeneous knowledge graph with node types *gene*, *pathway*, *drug*, *trial*, and *disease subtype*, modelled on the multilayer network-medicine approach validated for breast cancer drug repurposing ^[11] and on multi-layer guilt-by-association graph learning ^[10]. This graph is the substrate over which both the GNN synergy model (L3) and the retrieval layer used by the LLM agents (L4) operate, so a single curated source of truth replaces the previously separate database and ad hoc combination-enumeration logic.

4.2 Layer 3 — GNN Synergy Prediction

For every pair (and, where computationally tractable, triplet) of altered kinases with an approved or investigational inhibitor, L3 constructs a cancer-specific featured graph — the patient's altered genes and their PPI neighbourhood — and scores predicted synergy using a GNN trained in the style of SynerGNet ^[1] or DeepDDS ^[2], with the SANEpools attention-pooling mechanism ^[3] used to expose which sub-network drove each score, preserving interpretability. Where kinase-inhibitor and mTOR/DNA-damage/HDAC-inhibitor pairings are involved, the model additionally weights toward the mechanism classes previously shown to be consistently synergistic in FDA-approved drug screens ^[4].

4.3 Layer 4 — Multi-Agent LLM Reasoning

Rather than a single LLM narrative pass, L4 uses a small team of role-specialised agents, following the multi-agent MDT and evidence-based-medicine agent patterns described above:

- **Genomics Interpreter Agent** — summarises the patient's variant profile and activated pathways from L1/L2.
- **Pharmacology / Synergy Agent** — interprets the GNN's synergy scores (L3) and mechanistic rationale.
- **Clinical Evidence Agent (RAG)** — retrieves and summarises the strongest matching clinical-trial and PubMed evidence for each candidate combination, following the EBM-agent and MDT-RAG patterns ^[6,7].
- **Toxicity & Safety Agent** — screens candidate combinations for known overlapping organ toxicity or drug-drug interaction flags before they can be ranked highly.

- **Chair / Consensus Agent** — reconciles the other agents' outputs into the final ranked, six-section clinical narrative, resolving disagreements explicitly rather than silently averaging them, and flagging any combination on which agents disagree for mandatory human review.

This decomposition gives the system an internal adversarial check that a single LLM pass cannot provide: the Safety Agent can veto or down-rank a combination the Synergy Agent favours, and the Chair Agent must surface that disagreement to the clinician rather than resolve it silently.

4.4 Layer 5 — Adaptive Feedback (Optional, Future Work)

Once deployed with clinician-in-the-loop feedback, an RL module can treat each recommended ranking as an action and use acceptance/outcome signals as reward, in the spirit of RL-based chemotherapy dosing controllers [13,14,15]. This layer is proposed as future work rather than an initial requirement, since it depends on accumulated outcome data that a first deployment would not yet have.

5. Comparative Analysis

Dimension	Current Rule-Based Pipeline	Proposed MC-ORE Approach
Scoring basis	Shared pathway count (sigmoid heuristic)	Learned synergy prediction (GNN) + pathway context
Evidence grounding	Static, curated 10-gene database	Live knowledge graph + RAG over PubMed/trial registries
Reasoning	Single LLM narrative pass	Multi-agent deliberation with explicit disagreement flags
Toxicity awareness	Not modelled	Dedicated Safety Agent screens every candidate
Interpretability	Formula-based, transparent but coarse	Attention-based sub-network explanations (SANEpool-style)
Extensibility	Manual database expansion	Graph grows via automated KG ingestion pipelines
Adaptivity over time	None	Optional RL feedback layer (future work)
Privacy architecture	Local-first, de-identified summary only	Unchanged — preserved by design

6. Illustrative Application to the Sample Patient Profile

Applied to the same illustrative 9-kinase patient profile used to validate the original pipeline (EGFR, MET, ERBB2, SRC, ABL1, FGFR1–3, ALK; MAPK/ERK and PI3K/AKT redundancy scores of 0.865 and 0.848), the original rule-based analyzer enumerated 36 shared-pathway pairs and ranked five by literature-evidence strength, topped by EGFR+MET (erlotinib + crizotinib). MC-ORE would process the same 36 candidate pairs through the GNN synergy layer to obtain a *predicted synergy score* for each — not merely a shared-pathway count — and would

additionally evaluate selected higher-order triplets (e.g. EGFR+MET+SRC) that the original pipeline's pairwise-only design could not consider.

The Toxicity & Safety Agent would then screen the GNN's top-ranked candidates for overlapping adverse-effect profiles (for example, hepatotoxicity overlap between certain kinase-inhibitor pairs) before the Chair Agent finalises the ranked narrative — a screening step entirely absent from the original design. This example is illustrative of the architectural difference only; it is not a validated clinical result, and any real deployment would require retraining and validation on institution-specific data before such rankings could inform care.

Disclaimer: As with the original pipeline, this proposed system is intended strictly as a research and decision-support aid. It does not constitute diagnostic or prescriptive output and all recommendations require review by qualified oncology specialists and institutional tumour boards.

7. Evaluation and Validation Plan

4. **Synergy-model benchmarking:** Train and validate the L3 GNN against DrugCombDB and AZ-DREAM Challenge synergy labels before deployment, following the SynerGNet validation protocol ^[1], reporting balanced accuracy and generalisation to unseen drug/cell-line pairs.
5. **Retrospective case review:** Run MC-ORE on de-identified retrospective TNBC cases with known treatment outcomes and compare its ranked recommendations against the regimens actually selected by the treating tumour board.
6. **Multi-agent agreement analysis:** Measure how often the Safety Agent flags a combination the Synergy Agent ranks highly, and audit a sample of flagged cases with clinical pharmacists.
7. **Expert panel review:** Have oncologists and pharmacologists blind-score MC-ORE's explanations for clarity and clinical plausibility, comparable to expert-guided LLM evaluation frameworks reported in the oncology decision-support literature ^[8,9].

8. Ethical, Regulatory, and Governance Considerations

- The local-first, de-identified data architecture of the original pipeline is preserved: raw genomic data never leaves the researcher's or institution's environment.
- Every MC-ORE output remains advisory; a human-in-the-loop review requirement is enforced structurally, not just by disclaimer, via the Chair Agent's mandatory disagreement-flagging behaviour.
- Because the system would support real treatment decisions, any clinical pilot should be scoped and reviewed as a potential Software-as-a-Medical-Device (SaMD) under applicable regulatory frameworks before use beyond research settings.
- Knowledge-graph and RAG sources should be version-controlled and dated, so that every recommendation can be traced to the specific evidence snapshot that produced it.

9. Limitations

- GNN synergy models trained on cell-line data (e.g. DrugCombDB, AZ-DREAM) may not transfer perfectly to in vivo or patient-specific responses; prospective validation is required.

- Multi-agent LLM pipelines introduce additional latency and computational cost relative to a single-agent design.
- Knowledge-graph construction from clinical-trial text via LLM extraction (as in the network-medicine breast-cancer case study) requires ongoing quality control to avoid propagating extraction errors into downstream rankings.
- The optional RL feedback layer requires an outcome-data pipeline that does not yet exist and raises its own data-governance requirements.

10. Translational Validation: Closing the Wet-Lab Feedback Loop

A computational architecture is only as credible as the experimental loop that keeps it honest. The layers proposed in Section 4 (referred to here under their working translational-program names — the overall system as *Hybrid-CORE*, the Layer 2 knowledge graph as *KG++*, and the Layer 3 synergy model as *MM-SynergyNet*) only produce testable hypotheses; a wet-lab program is what confirms, refutes, and ultimately retrains them. The seven-stage loop below anchors the model to biology and, eventually, to clinician judgment.

Stage	Focus	Key Actions	Feeds
1	Define model-derived hypotheses	Extract top-ranked pairs/triplets with high synergy and acceptable safety scores; record predicted synergy score, tissue selectivity, and mechanistic subgraph; prioritise 10–20 combinations per indication.	L3, L4
2	Establish experimental models	Use patient-derived organoids or PDX models where possible, complemented by cell lines matching key pathway activations (e.g. MAPK/ERK, PI3K/AKT); include ≥ 1 non-malignant control line.	L1, L2
3	Run dose-response & synergy assays	Checkerboard dose-response matrices for pairs and triplets; measure viability, apoptosis, and pathway readouts (p-ERK, p-AKT); compute Bliss, Loewe, and HSA synergy scores in parallel rather than any single metric alone.	L3
4	Profile toxicity & off-target effects	Test on non-malignant organoids/primary cells (e.g. MCF10A, hepatocytes, cardiomyocytes) matched to the malignant model; measure cytotoxicity and organ-injury markers; compare directly against the Safety Agent's veto decisions.	L4
5	Map results back to KG++ / MM-SynergyNet	Annotate each tested combination with observed synergy, toxicity, and mechanistic plausibility; update KG++ edge weights where results confirm or refute bypass circuits; fine-tune MM-SynergyNet on new labelled data, weighted toward discrepant cases rather than a fixed retraining schedule.	L2, L3
6	Close the clinician feedback loop	Present validated combinations and mechanistic rationale to a tumour board/expert panel; capture acceptance, rejection, or modification as structured feedback; use these signals as the RL reward.	L5
7	Iterate and expand	Refresh the hypothesis set on a fixed cadence; extend to new	All

Stage	Focus	Key Actions	Feeds
		pathways/resistance phenotypes; track the in silico → in vitro → clinician-interest conversion rate as the program's core KPI.	layers

In steady state this produces a tight loop: **Hybrid-CORE proposes → wet lab tests in organoids/cell lines → synergy and toxicity are quantified → KG++ and MM-SynergyNet are updated → clinicians react to the most credible hypotheses → RL nudges future recommendations toward patterns that survive both biology and human judgment.** The critical design choice is Stage 6: without a structured clinician-feedback channel, the loop optimises for in vitro synergy alone, which is necessary but not sufficient for clinical relevance.

11. Concrete Experimental Design: EGFR + MET-Driven TNBC

Applying the loop above to the EGFR+MET candidate combination already surfaced in Section 6 (Layer 3's top-ranked pair for the illustrative 9-kinase patient profile) gives a first, indication-specific pilot:

11.1 Model Systems

- Primary model: an EGFR-amplified TNBC cell line (e.g. MDA-MB-468) plus, where available, a patient-derived TNBC organoid line with matching EGFR/MET co-alteration.
- Non-malignant control: MCF10A (immortalised, non-malignant breast epithelial line) run through an identical plate design.
- Systemic-toxicity proxy: primary hepatocytes, given known hepatotoxicity signals for several EGFR/MET inhibitor classes.

11.2 Plate Design & Readouts

- 8×8 checkerboard dose-response matrix for erlotinib (EGFR) × crizotinib (MET), half-log dilution series bracketing each drug's published IC50; n = 3 biological replicates.
- Viability via CellTiter-Glo at 72h; apoptosis via Annexin V/caspase-3/7; pathway engagement via phospho-EGFR, phospho-MET, phospho-ERK, and phospho-AKT (Western blot or Luminex multiplex).
- Synergy quantified in SynergyFinder using Bliss independence, Loewe additivity, and HSA in parallel; a combination is only advanced if ≥ 2 of 3 metrics agree on synergy.

Optional triplet arm

If the pairwise EGFR+MET signal is confirmed, a reduced-matrix triplet arm adding an SRC inhibitor (informed by the pipeline's MAPK/ERK redundancy score) tests whether a third agent improves synergy without compounding toxicity beyond the Safety Agent's threshold.

11.3 Data Flow Back Into the Five Layers

- Viability/synergy data → MM-SynergyNet fine-tuning set (L3), weighted toward any case where predicted and observed synergy disagree.
- Toxicity data on MCF10A/hepatocytes → direct calibration check on the Safety Agent's veto threshold (L4).

- Confirmed or refuted mechanistic subgraphs (e.g. an EGFR→MET bypass edge) → KG++ edge-weight update (L2).
- Tumour-board reaction to the EGFR+MET(+SRC) narrative → structured RL reward signal (L5).

A single indication-specific cycle of this kind (Stages 1–6 above) is estimated at roughly 8–12 weeks from hypothesis extraction to tumour-board review, which sets a realistic cadence for Stage 7's iterate-and-expand step across additional indications.

12. Conclusion

Replacing a single rule-based redundancy score and one-pass LLM synthesis with a learned GNN synergy model, a knowledge-graph-grounded retrieval layer, and a multi-agent LLM reasoning process addresses the three most consequential gaps in the original tnbc-genomics-agent design: synergy is predicted rather than assumed from pathway overlap, safety is explicitly screened rather than ignored, and recommendations are produced through an auditable, adversarially-checked reasoning process rather than a single narrative pass. Every component of the proposed MC-ORE architecture is grounded in published, peer-reviewed methods already demonstrated in oncology or in breast cancer specifically, making this a realistic and incremental — rather than speculative — extension of the existing pipeline.

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Background context on TNBC biology, kinase frequencies, and the original pipeline architecture is drawn from the companion report, "TNBC Genomics Agent: RTK/nRTK Redundancy Analysis Pipeline" (v1.0.0, June 2026), which should be read alongside this proposal.